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Approximating discrete probability distributions with dependence trees

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Abstract

A method is presented to approximate optimally an n-dimensional discrete probability distribution by a product of second-order distributions, or the distribution of the first-order tree dependence. The problem is to find an optimum set of n - 1 first order dependence relationship among the n variables. It is shown that the procedure derived in this paper yields an approximation of a minimum difference in information. It is further shown that when this procedure is applied to empirical observations from an unknown distribution of tree dependence, the procedure is the maximum-likelihood estimate of the distribution.

Real-World Impact & Application

Foundations of Probabilistic Models & AI: The Chow-Liu tree algorithm proved that complex multi-variable probability distributions can be optimally approximated as a tree of mutual information. Today, this fundamental insight is used across speech processing, computer vision, Bayesian belief networks, and modern NLP dependency parsers to capture feature dependencies efficiently without exponential compute costs.